SC1 (22 nF) Data Table

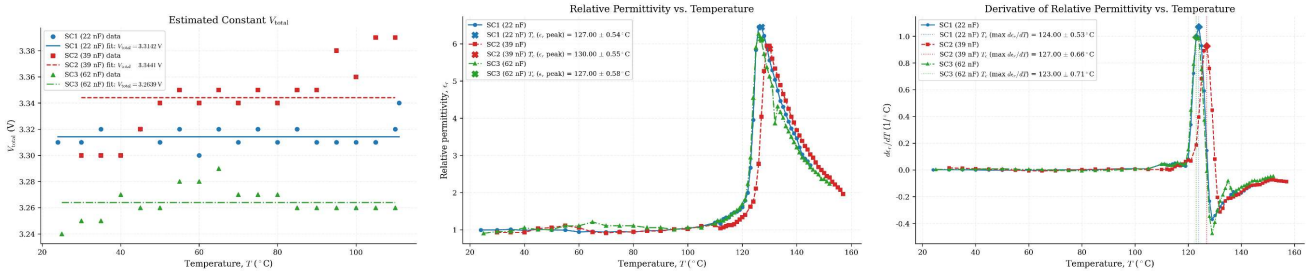
Temperature (°C)	V_{dc} (V)	V_{sc} (V)	$C = V_{\mathrm{sc}}/V_{\mathrm{dc}}$	ϵ_r	$d\epsilon_r/dT$ (1/°C)
24.0	2.67	0.64	0.2397	0.998304	0.001159
30.0	2.67	0.64	0.2397	0.998304	0.000886
35.0	2.67	0.65	0.243446	1.013902	0.000291
40.0	2.67	0.63	0.235955	0.982705	-0.000442
45.0	2.67	0.65	0.243446	1.013902	-0.001227
50.0	2.67	0.64	0.2397	0.998304	-0.002291
55.0	2.68	0.64	0.238806	0.994579	-0.002375
60.0	2.69	0.61	0.226766	0.944434	-0.002722
65.0	2.7	0.62	0.22963	0.956361	-0.002382
70.0	2.7	0.61	0.225926	0.940936	-0.00071
75.0	2.7	0.62	0.22963	0.956361	0.001014
80.0	2.7	0.61	0.225926	0.940936	0.002051
85.0	2.69	0.63	0.234201	0.975399	0.002472
90.0	2.68	0.63	0.235075	0.979038	0.004761
95.0	2.66	0.65	0.244361	1.017714	0.008801
100.0	2.65	0.66	0.249057	1.037271	0.009261
105.0	2.62	0.69	0.263359	1.096836	0.008137
110.0	2.58	0.74	0.286822	1.194555	0.041687
111.0	2.58	0.76	0.294574	1.22684	0.043492
112.0	2.59	0.724211	0.279618	1.164553	0.036924
113.0	2.53	0.784211	0.309965	1.29094	0.036999
114.0	2.51	0.804211	0.320403	1.334412	0.044683
115.0	2.51	0.804211	0.320403	1.334412	0.052377
116.0	2.47	0.844211	0.341786	1.423468	0.039849
117.0	2.45	0.864211	0.352739	1.469087	0.033654
118.0	2.44	0.874211	0.358283	1.492177	0.035183
119.0	2.42	0.894211	0.369508	1.538928	0.028297
120.0	2.39	0.924211	0.386699	1.610523	0.082147
121.0	2.32	0.994211	0.428539	1.784779	0.33809
122.0	2.24	1.074211	0.479558	1.997264	0.723233
123.0	2.02	1.294211	0.640698	2.668379	0.998798
124.0	1.7	1.614211	0.949536	3.954625	1.069986
125.0	1.38	1.934211	1.401602	5.837389	0.929063
126.0	1.3	2.014211	1.549393	6.452909	0.591785
127.0	1.3	2.014211	1.549393	6.452909	0.14512
128.0	1.33	1.984211	1.491888	6.213411	-0.233724
129.0	1.37	1.944211	1.419132	5.910398	-0.369384
130.0	1.42	1.894211	1.333951	5.555638	-0.341726
131.0	1.46	1.854211	1.270007	5.289324	-0.309399
132.0	1.5	1.814211	1.209474	5.037214	-0.275468
133.0	1.55	1.764211	1.1382	4.740375	-0.246735
134.0	1.6	1.714211	1.071382	4.462089	-0.229586
135.0	1.63	1.684211	1.033258	4.303311	-0.21482
136.0	1.67	1.644211	0.984557	4.100482	-0.184148
137.0	1.71	1.604211	0.938135	3.907143	-0.16281
138.0	1.75	1.564211	0.893835	3.722641	-0.170776
139.0	1.78	1.534211	0.861916	3.589707	-0.179344
140.0	1.81	1.504211	0.831056	3.461179	-0.172576
141.0	1.87	1.444211	0.772305	3.216495	-0.156537
142.0	1.92	1.394211	0.726151	3.024274	-0.140264
143.0	1.95	1.364211	0.699595	2.913673	-0.128642
144.0	1.97	1.344211	0.68234	2.84181	-0.11702
145.0	2.0	1.314211	0.657105	2.736711	-0.111209

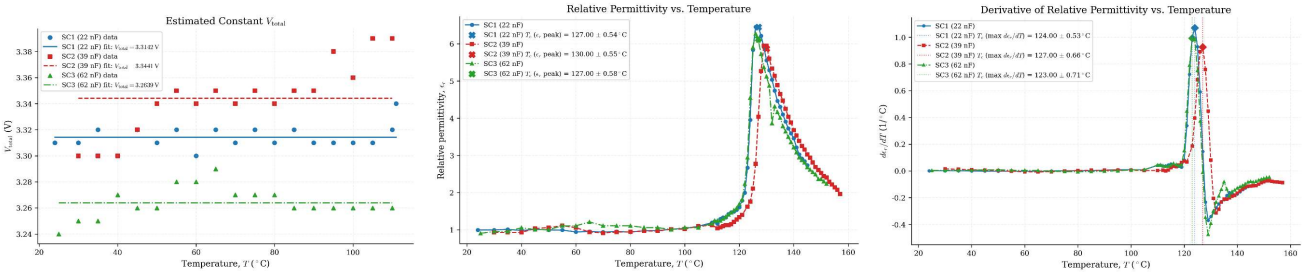
SC2 (39 nF) Data Table

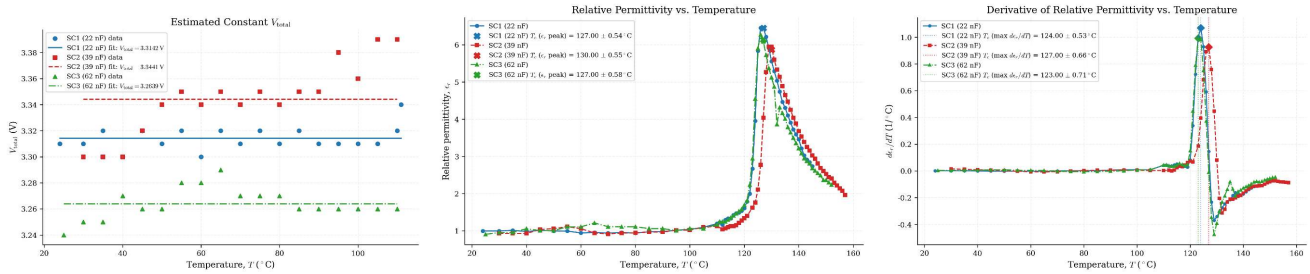
Temperature (°C)	V_{dc} (V)	V_{sc} (V)	$C = V_{\mathrm{sc}}/V_{\mathrm{dc}}$	ϵ_r	$d\epsilon_r/dT$ (1/°C)
30.0	2.93	0.37	0.12628	0.929929	0.014456
35.0	2.93	0.37	0.12628	0.929929	0.012544
40.0	2.93	0.37	0.12628	0.929929	0.008719
45.0	2.91	0.41	0.140893	1.037544	0.004895
50.0	2.92	0.42	0.143836	1.05921	0.001581
55.0	2.91	0.44	0.151203	1.113461	-0.002042
60.0	2.92	0.42	0.143836	1.05921	-0.006688
65.0	2.97	0.38	0.127946	0.942199	-0.007924
70.0	2.97	0.37	0.124579	0.917404	-0.006039
75.0	2.97	0.38	0.127946	0.942199	-0.001845
80.0	2.96	0.38	0.128378	0.945382	0.003011
85.0	2.96	0.39	0.131757	0.970261	0.004587
90.0	2.96	0.39	0.131757	0.970261	0.006184
95.0	2.97	0.41	0.138047	1.016583	0.006614
100.0	2.95	0.41	0.138983	1.023475	0.004588
105.0	2.95	0.44	0.149153	1.098364	0.003017
110.0	2.94	0.45	0.153061	1.127147	0.003191
112.0	2.93	0.414118	0.141337	1.04081	0.004467
113.0	2.92	0.424118	0.145246	1.069594	0.001333
114.0	2.91	0.434118	0.149181	1.098576	0.00601
115.0	2.9	0.444118	0.153144	1.127757	0.025312
116.0	2.9	0.444118	0.153144	1.127757	0.034995
117.0	2.89	0.454118	0.157134	1.15714	0.031322
118.0	2.87	0.474118	0.165198	1.216521	0.04336
119.0	2.85	0.494118	0.173375	1.276736	0.06385
120.0	2.83	0.514118	0.181667	1.337801	0.073482
121.0	2.81	0.534118	0.190077	1.399736	0.067281
123.0	2.74	0.604118	0.220481	1.623628	0.185304
124.0	2.7	0.644118	0.238562	1.756778	0.395957
125.0	2.6	0.744118	0.286199	2.107578	0.682155
126.0	2.43	0.914118	0.37618	2.770201	0.892982
127.0	2.16	1.184118	0.548203	4.03698	0.926449
128.0	1.95	1.394118	0.714932	5.264781	0.75957
129.0	1.85	1.494118	0.807631	5.947419	0.446522
130.0	1.85	1.494118	0.807631	5.947419	0.081938
131.0	1.9	1.444118	0.760062	5.597118	-0.20487
132.0	1.94	1.404118	0.723772	5.329878	-0.314034
133.0	1.97	1.374118	0.697522	5.136569	-0.287973
134.0	2.01	1.334118	0.66374	4.887801	-0.233725
135.0	2.05	1.294118	0.631277	4.648741	-0.21124
136.0	2.08	1.264118	0.607749	4.47548	-0.2101
137.0	2.12	1.224118	0.577414	4.252093	-0.202866
138.0	2.16	1.184118	0.548203	4.03698	-0.189663
139.0	2.19	1.154118	0.526994	3.880801	-0.182624
140.0	2.23	1.114118	0.499604	3.6791	-0.167302
141.0	2.26	1.084118	0.479698	3.53251	-0.148304
142.0	2.29	1.054118	0.460313	3.389761	-0.135851
143.0	2.32	1.024118	0.44143	3.250703	-0.120791
144.0	2.34	1.004118	0.42911	3.159979	-0.114321
145.0	2.37	0.974118	0.41102	3.026764	-0.114374
146.0	2.39	0.954118	0.399212	2.939812	-0.110762
147.0	2.42	0.924118	0.381867	2.812078	-0.109416
148.0	2.45	0.894118	0.364946	2.687473	-0.099274
149.0	2.47	0.874118	0.353894	2.606084	-0.087029
150.0	2.49	0.854118	0.343019	2.526003	-0.080176
151.0	2.51	0.834118	0.332318	2.447198	-0.073892
152.0	2.52	0.824118	0.327031	2.408264	-0.073435
153.0	2.55	0.794118	0.311419	2.293296	-0.076976
154.0	2.57	0.774118	0.301213	2.218142	-0.081189
155.0	2.59	0.754118	0.291165	2.144148	-0.08357
156.0	2.61	0.734118	0.281271	2.071289	-0.085952
157.0	2.64	0.704118	0.266711	1.964069	-0.087142

SC3 (62 nF) Data Table

Temperature (°C)	V_{dc} (V)	V_{sc} (V)	$C = V_{\mathrm{sc}}/V_{\mathrm{dc}}$	ϵ_r	$d\epsilon_r/dT$ (1/°C)
25.0	3.06	0.18	0.058824	0.906475	0.003364
30.0	3.06	0.19	0.062092	0.956835	0.003802
35.0	3.06	0.19	0.062092	0.956835	0.004679
40.0	3.06	0.21	0.068627	1.057554	0.005555
45.0	3.06	0.2	0.065359	1.007194	0.00697
50.0	3.06	0.2	0.065359	1.007194	0.007461
55.0	3.06	0.22	0.071895	1.107914	0.005362
60.0	3.06	0.22	0.071895	1.107914	0.004804
65.0	3.05	0.24	0.078689	1.212596	0.003099
70.0	3.05	0.22	0.072131	1.111546	-0.001704
75.0	3.05	0.22	0.072131	1.111546	-0.004942
80.0	3.05	0.22	0.072131	1.111546	-0.006505
85.0	3.05	0.21	0.068852	1.061021	-0.005037
90.0	3.05	0.21	0.068852	1.061021	-0.00259
95.0	3.06	0.2	0.065359	1.007194	0.001109
100.0	3.05	0.21	0.068852	1.061021	0.006711
105.0	3.05	0.21	0.068852	1.061021	0.009216
110.0	3.03	0.23	0.075908	1.169741	0.043994
111.0	3.02	0.243889	0.080758	1.244485	0.044753
112.0	3.02	0.243889	0.080758	1.244485	0.039184
113.0	3.02	0.243889	0.080758	1.244485	0.034667
114.0	3.01	0.253889	0.084348	1.299816	0.035384
115.0	3.0	0.263889	0.087963	1.355516	0.04684
116.0	2.99	0.273889	0.091602	1.411588	0.056006
117.0	2.98	0.283889	0.095265	1.468036	0.048317
118.0	2.97	0.293889	0.098952	1.524865	0.040426
119.0	2.96	0.303889	0.102665	1.582078	0.045343
120.0	2.94	0.323889	0.110166	1.69767	0.152217
121.0	2.92	0.343889	0.11777	1.814847	0.4478
122.0	2.85	0.413889	0.145224	2.237915	0.790844
123.0	2.74	0.523889	0.1912	2.946411	0.993279
124.0	2.52	0.743889	0.295194	4.548961	0.981117
125.0	2.36	0.903889	0.383004	5.902116	0.751092
126.0	2.32	0.943889	0.406849	6.269567	0.373172
127.0	2.34	0.923889	0.394824	6.084271	-0.010796
128.0	2.38	0.883889	0.371382	5.723022	-0.34731
129.0	2.42	0.843889	0.348714	5.373714	-0.474501
130.0	2.45	0.813889	0.3322	5.119219	-0.390772
131.0	2.48	0.783889	0.316084	4.870881	-0.299157
132.0	2.61	0.653889	0.250532	3.860718	-0.231899
133.0	2.55	0.713889	0.279956	4.314149	-0.185955
134.0	2.57	0.693889	0.269996	4.160653	-0.136646
135.0	2.59	0.673889	0.260189	4.009528	-0.08089
136.0	2.62	0.643889	0.245759	3.787166	-0.135633
137.0	2.64	0.623889	0.236322	3.641732	-0.187133
138.0	2.66	0.603889	0.227026	3.498485	-0.153686
139.0	2.68	0.583889	0.217869	3.357377	-0.14283
140.0	2.7	0.563889	0.208848	3.218359	-0.127896
141.0	2.72	0.543889	0.199959	3.081385	-0.118011
142.0	2.74	0.523889	0.1912	2.946411	-0.108682
143.0	2.75	0.513889	0.186869	2.87966	-0.096847
144.0	2.76	0.503889	0.182568	2.813393	-0.082523
145.0	2.78	0.483889	0.174061	2.682289	-0.079031
146.0	2.79	0.473889	0.169853	2.617442	-0.08299
147.0	2.8	0.463889	0.165675	2.553058	-0.077129
148.0	2.81	0.453889	0.161526	2.489132	-0.068863
149.0	2.83	0.433889	0.153318	2.362636	-0.062384
150.0	2.83	0.433889	0.153318	2.362636	-0.055625
151.0	2.84	0.423889	0.149257	2.300056	-0.048865
152.0	2.85	0.413889	0.145224	2.237915	-0.045485


Estimated Constant V_{total}
SC1 (22 nF) data
SC1 (22 nF) fit: $V_{\text{total}} = 3.3142$ V
SC2 (39 nF) data
SC2 (39 nF) fit: $V_{\text{total}} = 3.3441$ V
SC3 (62 nF) data
SC3 (62 nF) fit: $V_{\text{total}} = 3.2639$ V
V_{total} (V)
Temperature, T (°C)
Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (ϵ_r peak) = 127.00 ± 0.54 °C
SC2 (39 nF)
SC2 (39 nF) T_c (ϵ_r peak) = 130.00 ± 0.55 °C
SC3 (62 nF)
SC3 (62 nF) T_c (ϵ_r peak) = 127.00 ± 0.58 °C
Relative permittivity, ϵ_r
Temperature, T (°C)
Derivative of Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (max $d\epsilon_r/dT$) = 124.00 ± 0.53 °C
SC2 (39 nF)
SC2 (39 nF) T_c (max $d\epsilon_r/dT$) = 127.00 ± 0.66 °C
SC3 (62 nF)
SC3 (62 nF) T_c (max $d\epsilon_r/dT$) = 123.00 ± 0.71 °C
$d\epsilon_r/dT$ (1/°C)
Temperature, T (°C)


Estimated Constant V_{total}
SC1 (22 nF) data
SC1 (22 nF) fit: $V_{\text{total}} = 3.3142$ V
SC2 (39 nF) data
SC2 (39 nF) fit: $V_{\text{total}} = 3.3441$ V
SC3 (62 nF) data
SC3 (62 nF) fit: $V_{\text{total}} = 3.2639$ V
V_{total} (V)
Temperature, T (°C)
Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (ϵ_r peak) = 127.00 ± 0.54 °C
SC2 (39 nF)
SC2 (39 nF) T_c (ϵ_r peak) = 130.00 ± 0.55 °C
SC3 (62 nF)
SC3 (62 nF) T_c (ϵ_r peak) = 127.00 ± 0.58 °C
Relative permittivity, ϵ_r
Temperature, T (°C)
Derivative of Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (max $d\epsilon_r/dT$) = 124.00 ± 0.53 °C
SC2 (39 nF)
SC2 (39 nF) T_c (max $d\epsilon_r/dT$) = 127.00 ± 0.66 °C
SC3 (62 nF)
SC3 (62 nF) T_c (max $d\epsilon_r/dT$) = 123.00 ± 0.71 °C
$d\epsilon_r/dT$ (1/°C)
Temperature, T (°C)


Estimated Constant V_{total}
SC1 (22 nF) data
SC1 (22 nF) fit: $V_{\text{total}} = 3.3142$ V
SC2 (39 nF) data
SC2 (39 nF) fit: $V_{\text{total}} = 3.3441$ V
SC3 (62 nF) data
SC3 (62 nF) fit: $V_{\text{total}} = 3.2639$ V
V_{total} (V)
Temperature, T (°C)
Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (ϵ_r peak) = 127.00 ± 0.54 °C
SC2 (39 nF)
SC2 (39 nF) T_c (ϵ_r peak) = 130.00 ± 0.55 °C
SC3 (62 nF)
SC3 (62 nF) T_c (ϵ_r peak) = 127.00 ± 0.58 °C
Relative permittivity, ϵ_r
Temperature, T (°C)
Derivative of Relative Permittivity vs. Temperature
SC1 (22 nF)
SC1 (22 nF) T_c (max $d\epsilon_r/dT$) = 124.00 ± 0.53 °C
SC2 (39 nF)
SC2 (39 nF) T_c (max $d\epsilon_r/dT$) = 127.00 ± 0.66 °C
SC3 (62 nF)
SC3 (62 nF) T_c (max $d\epsilon_r/dT$) = 123.00 ± 0.71 °C
$d\epsilon_r/dT$ (1/°C)
Temperature, T (°C)